



UNIVERSITY
of HULL

Computer Science

BSc / MEng Computer Science

October 2026

Computer Science

Shaded modules are offered as options on your programme, all other modules are compulsory.

Level 4 (Certificate Stage)

Trimester 1	Trimester 2
Programming Portfolio (40cr. core)	
Algorithms and Data Structures	Databases and the Cloud
Professional Development	Computational Thinking

Common first year allows for easy transfer between the available programmes.

Level 5 (Diploma Stage)

Trimester 1	Trimester 2
Design, Develop, Deploy (40cr. core)	
Artificial Intelligence	Distributed Systems Programming
Foundations of Cybersecurity and Secure Development	Advanced Programming
Graphics Programming	eXtended Reality Development
Electronics and Microcontrollers	Data Analysis and Visualisation
	Connected Devices and IoT

Post-Diploma Stage (optional)

Trimester 1	Trimester 2
Year in Industry/Year Abroad	

Level 6 (Honours Stage)

Trimester 1	Trimester 2
Honours Stage Project (40 credits)	
Communicating and Teaching Computing (40 credits)	
Computer Vision	Natural Language Processing
Secure Digital Infrastructure	Safety Critical Systems
Digital Disruption and Innovation	Generative and Collaborative AI
Machine Learning	Embedded Systems Applications
Network Security	Parallel and Concurrent programming
Simulation and Artificial Intelligence for Games Programming	Ethical Hacking
Games Architecture	

Level 7 (Masters Stage) (optional)

Trimester 1	Trimester 2
Commercial Development Practice (40 credits)	
Dependable Reactive Real-Time Systems	Cutting Edge Concepts
Component Based Software Engineering Practice	Trustworthy Computing
Real-time Graphics	Responsible, Safe and Explainable AI
C++ Programming and Design	Simulation and Concurrency
Edge AI	
Cyber Defence and Threat Intelligence	

All of the modules are available to be selected as options, however they must be taken in the pairs prescribed in the specific programme variants to ensure that the programme competencies are met.

Pairings for trimester 1 are as follows:

- Dependable Reactive Real-Time Systems *and* Component Based Software Engineering Practice, or
- C++ Programming and Design *and* Real-time Graphics, or
- Dependable Reactive Real-Time Systems *and* Edge AI, or
- Cyber Defence and Threat Intelligence *and* Component Based Software Engineering Practice

Cutting Edge Concepts is compulsory in trimester 2. A single module must additionally be chosen from the following options:

- Trustworthy Computing, or
- Responsible, Safe and Explainable AI or
- Simulation and Concurrency, or
 - Must have taken the C++ Programming and Design with Real-time Graphics pair in Trimester 1.

Please note that the content in the following pages is subject to change, as our curriculum evolves to match the rapidly changing Computer Science landscape.

Level 4

Programming Portfolio

Core tools: C#

Whether you've never written a line of code or you've been tinkering for years, this module meets you where you are. You'll learn to think like a programmer — breaking problems down, writing clean logic, and building working software in C#. By the end of the year you'll have moved from your very first program to designing with classes, objects and inheritance.

You'll explore:

- Core programming building blocks — variables, loops, functions and data types
- Searching, sorting and working with arrays, stacks and queues
- Object-oriented programming — classes, inheritance and encapsulation
- Reading, writing and managing code like a professional, using source control

Algorithms and Data Structures

Every great piece of software starts with a great idea for how to solve the problem — that's what this module teaches you to find. You'll step away from the screen and learn to think in code

before you write it, exploring the algorithms and data structures that separate software that merely works from software that works brilliantly.

You'll explore:

- Pseudocode and state machines for planning before you program
- Core data structures, and how to choose the right one for the job
- Sorting and searching algorithms, and how to measure their efficiency
- Design thinking that carries into every other module on your degree

Databases and the Cloud

Core tools: C#, Microsoft SQL Server

Modern software rarely stores its data on just one machine anymore — this module shows you how databases and the cloud work together to make that possible. You'll learn how cloud computing and virtualisation actually work, then design and build real databases, from relational systems through to NoSQL, before bringing the two together to build a working application backed by a database hosted in the cloud.

You'll explore:

- Designing and implementing relational and NoSQL databases
- Virtualisation, containers, and the cloud service models behind them — IaaS, PaaS and SaaS
- Connecting a real application to a cloud-hosted database
- Cloud security, reliability and disaster recover

Computational Thinking

Core tools: programming platforms and mathematical software

Computing is bigger than any one programming language — this module is about how to think as a computer scientist. Drawing on a genuinely global history of innovation, you'll build the mathematical and logical foundations that everything else on your degree rests on.

You'll explore:

- Abstraction and computational methods
- The maths behind computing — algebra, number systems and statistics
- Logic, Boolean algebra and set theory
- Vectors and matrices, and their uses in graphics, AI and data
- Practical aspects of how computers handle different number types and the way that mathematical libraries and tools can help to do the actual calculations

Professional Development

A computer science degree is about more than code — this module helps you build the career to go with it. You'll explore the ethical, legal, social and sustainability challenges facing the

sector, and the professional skills that set graduates apart. A hands-on team project puts those skills into practice straight away, while you start building a portfolio that showcases your work.

You'll explore:

- Sector challenges — ethics, diversity, sustainability and cybersecurity
- Communication, research and self-presentation skills
- Real teamwork on a programme-specific group project
- Building your own professional portfolio

Level 5

Design, Develop, Deploy

Core tools: GitHub, GitHub Actions

This is where you learn to build software the way the industry actually builds it — from first idea to live deployment. As the backbone module of your second year, you'll work through the entire software development lifecycle, solo and as part of a team, picking up the methodologies and tools employers expect you to know.

You'll explore:

- Agile, iterative and waterfall development approaches
- Gathering and modelling real-world requirements
- Test-driven and behaviour-driven development
- Version control, CI/CD and cloud deployment with GitHub Actions

Advanced Programming

Core tools: C++

C++ powers everything from games engines to spacecraft — this module shows you why, from the inside out. You'll push beyond the basics of object-oriented programming and get close to the machine, exploring what's really happening between your code and the processor.

You'll explore:

- Templates and the Standard Template Library
- Composition and inheritance in advanced data structures
- Pointers, memory management and the “gang of three”
- Assembly language and debugging at the processor level

Artificial Intelligence

Learn to build the kind of intelligent systems that are reshaping every industry and understand the ethics behind them. You'll get a practical grounding in how AI actually works, from the history of the field to the algorithms powering it today, building your own solutions to real problem-solving challenges along the way.

You'll explore:

- The history and ethics and trustworthiness of artificial intelligence
- Pathfinding and problem-solving search algorithms
- Supervised and unsupervised learning, and neural networks
- Reasoning under uncertainty with Bayesian methods, and an introduction to reinforcement learning
- A brief introduction of fairness, explainability and AI Safety

Data Analysis and Visualisation

Data is only powerful once someone can make sense of it — this module teaches you to be that person. You'll work with real-world data from cleaning and wrangling it through to building and evaluating models, finishing with the skills to turn numbers into a story that's clear, compelling and honest.

You'll explore:

- Data cleaning, exploration and dimensionality reduction with pandas, scikit-learn and NumPy
- Classification, regression and clustering algorithms
- Rigorous experimentation, evaluation and hyperparameter tuning
- Spotting and mitigating bias in data and models
- Visualising data for maximum impact

Distributed Systems Programming

Core tools: .NET Core

Modern software rarely runs on just one machine — learn to build systems that work seamlessly across many. You'll design and build real distributed applications, navigating the unique challenges of concurrency, scalability, performance, security and scale that come with connecting devices and sharing workloads across a network.

You'll explore:

- Architectures and strategies for distributed computing
- Concurrency, security, performance and reliability
- Key distributed design patterns and best practice
- Deploying layered, scalable distributed applications

Electronics and Microcontrollers

Software needs hardware to run on — this module teaches you to build that hardware yourself. From first principles of voltage and current through to programming your own microcontroller, you'll get genuinely hands-on, building and testing real circuits on a breadboard.

You'll explore:

- Circuits, resistance, capacitors and semiconductors

- Microcontroller programming with Arduino
- Sensors and communication protocols like I2C and SPI
- Practical, hands-on electronic construction

Connected Devices and IoT

From smart homes to industrial sensors, the Internet of Things is everywhere — learn to build the systems that connect it all. You'll explore the communication technologies that let devices talk to each other and to the cloud, and learn to choose the right one for a given real-world application.

You'll explore:

- Wired and wireless interfacing technologies, from Bluetooth to Ethernet
- Publish/subscribe architectures and real-time communication
- Edge and cloud computing for connected devices
- Security and business considerations for IoT systems

Foundations of Cybersecurity and Secure Development

Cybersecurity is at the heart of today's digital world. This module provides a broad introduction to the principles, technologies, and practices that protect systems, networks, software, and data from cyber threats. Students will explore how security is integrated into the design and development of digital solutions and gain an understanding of the technical, organisational, and ethical aspects of cybersecurity. The module establishes the essential knowledge required for more advanced study in cybersecurity throughout the degree.

You'll explore:

- Fundamental cybersecurity concepts and security principles
- Cyber threats, vulnerabilities, and risk management approaches
- Secure system and software development practices
- Data protection fundamentals
- Governance, ethics, and legal aspects of cybersecurity
- Emerging trends and technologies in cybersecurity

Graphics Programming

Core tools: Shaders

Ever wondered how a game engine turns numbers into the world on your screen? This module shows you. Using a professional, industry-standard game engine, you'll learn the mathematics and pipeline behind real-time 3D graphics, and get hands-on writing your own shaders.

You'll explore:

- The evolution from fixed-function to programmable graphics pipelines
- Coordinate systems, vectors, matrices and transformations

- Building and animating 3D models and camera systems
- Writing custom shaders and using frame debugging tools

Year in Industry/Year Abroad

Take everything you've learned so far and put it to work — in a real workplace or a different country altogether. Spend a year gaining genuine professional experience on an industry placement, or broaden your horizons studying at an overseas institution. Either route builds transferable, employability-boosting skills in a credit-bearing year that counts towards your degree.

You'll explore:

- A year of real workplace experience, tailored to your placement
- Or a year studying abroad, guided by department staff to match your goals

Level 6

Honours Stage Project

This is your chance to lead a project entirely your own — from first idea to final report. You'll choose your own topic and see it through from planning to delivery, supported by videos, live sessions and workshops covering everything you need to succeed.

You'll explore:

- Project planning and literature review
- Experimental design, testing and evaluation
- Research ethics and data protection
- Professional report writing and presentation

Communicating and Teaching Computing

Think you understand computer science? Try explaining it to someone else — that's the real test. Spend real time in schools or colleges as a computing ambassador, supported by workshops on teaching practice and the National Curriculum, and discover how explaining your subject deepens your own understanding of it.

You'll explore:

- Classroom management and lesson planning
- How computer science is taught in schools today
- Reflective, experience-based learning from your own placement
- Effective communication skills – a critical life skill whatever career you choose

Generative and Collaborative AI

AI isn't going to replace you — but someone who knows how to work with it might. This module makes sure that's you. You'll build three essential “fusion skills” for working alongside

generative AI — asking it the right questions, judging its outputs critically, and teaching it to improve — then apply them to a real project of your own design.

You'll explore:

- Prompting techniques, from chain-of-thought to few-shot learning
- Critically evaluating AI-generated outputs for bias and accuracy
- Ethical, responsible use of generative AI
- A final project building a real generative AI solution

Computer Vision

Teach a computer to see — and to actually understand what it's looking at. From basic image processing through to recognising objects and interpreting entire scenes, you'll build the skills behind everything from facial recognition to self-driving cars.

You'll explore:

- Image processing and feature extraction
- Segmentation, shape analysis and motion tracking
- Object and activity recognition
- The real-world applications and ethics of computer vision

Digital Disruption and Innovation

Got an idea that could change an industry? This module teaches you how to turn it into a business. You'll learn the techniques real tech start-ups use to launch digital products, from lean methodology and business modelling to pitching for funding, all grounded in legal and ethical best practice.

You'll explore:

- The Lean Startup method and Business Model Canvas
- Digital marketing, branding and intellectual property
- Spotting disruption through technology horizon scanning
- Pitching innovative digital solutions

Embedded Systems Applications

Take a real physical system and give it a brain — this module is as hands-on as computer science gets. You'll design, build and deploy control logic for a genuine embedded system, working at the point where hardware and software meet.

You'll explore:

- Embedded processor architecture and assembly programming
- Hardware-software integration and peripheral interfacing
- Designing control systems, from bang-bang to PID controllers
- Building a complete real-time embedded control system

Ethical Hacking

Ethical hackers play a vital role in helping organisations identify and address security weaknesses before cybercriminals can exploit them. Through immersive, hands-on labs and realistic scenarios, you'll gain practical experience in penetration testing, vulnerability assessment, and security analysis while developing the professional and ethical mindset expected of modern cybersecurity practitioners.

You'll explore:

- Penetration testing methods and industry-standard security tools
- Network, wireless, and web application security
- Vulnerability discovery, analysis, and remediation
- AI-enhanced approaches to security testing and defence
- Professional penetration testing reporting and communication
- Legal, ethical, and regulatory aspects of ethical hacking

Games Architecture

Behind every great game is a great engine — learn how to build one. You'll explore the software architectures that power modern game engines, from component-based systems like Entity-Component-System through to the hardware considerations that shape how games run on console, PC and mobile.

You'll explore:

- Comparing and evaluating game engine architectures
- Entity-Component-System and modular design
- CPU architecture, bottlenecks and performance
- The asset pipeline and platform considerations

Machine Learning

Move beyond using machine learning models to really understanding how they work. Building on your foundational AI knowledge, you'll dig into the algorithms behind modern machine learning, from classic techniques through to the convolutional networks and generative AI shaping the field today.

You'll explore:

- Gradient descent, regression and regularisation
- Decision trees, random forests and clustering
- Convolutional neural networks
- An introduction to generative AI

Natural Language Processing

Teach a machine to read, understand and write human language — then push it further than you thought possible. You'll explore the techniques behind everything from chatbots to translation tools, working with the large language models and transformer architectures driving the field right now.

You'll explore:

- Large language models and transformer networks
- Generative and conversational AI
- Information retrieval and question answering
- Bias and fairness in language models

Parallel and Concurrent Programming

Core tools: CUDA, Rust

One processor is never enough — learn to make hundreds of them work together. You'll get hands-on with parallel programming on both CPU and GPU, implementing real algorithms in two very different languages to understand the challenges and rewards of true concurrency.

You'll explore:

- Shared memory vs message passing architectures
- Task, data and implicit parallelism
- Threads, channels and asynchronous programming
- Synchronisation, and avoiding the pitfalls of concurrency

Safety Critical Systems

When software failure means lives are at risk, 'good enough' isn't good enough — this module shows you what is. From aircraft to nuclear power stations, you'll explore the techniques used to analyse, model and mitigate risk in systems where failure simply isn't an option — including the unique safety challenges of AI and machine learning.

You'll explore:

- Fault Tree Analysis and Reliability Block Diagrams
- Markov Chains and Bayesian Networks for safety modelling
- Safety challenges unique to machine learning and artificial intelligence
- Real research applications, from drones to robotics

Secure Digital Infrastructure

Keeping the lights on for a business's digital systems is a job that never really stops — this module shows you what it takes. You'll learn to deploy, secure and maintain the infrastructure modern business runs on, from databases and email to disaster recovery.

You'll explore:

- Secure configuration of core business services
- Risk management and disaster recovery
- Systems and capacity monitoring
- Core infrastructure services — DNS, backup and virtualisation

Simulation and Artificial Intelligence for Games Programming

Give game characters a mind of their own — this module teaches you how. You'll write the data-driven code that brings games to life, building physics simulations and AI systems used in real production pipelines.

You'll explore:

- Physics-based simulation and collision response
- Pathfinding with Dijkstra maps and A*
- Steering behaviours, state machines and behaviour trees
- Advanced topics like flocking and procedural generation

Network Security

Core tools: Python, C++, C#, Wireshark

Every network is a target — this module teaches you to design, defend and probe them, the right way. You'll build a solid foundation in how networks actually work, from addressing and routing through to the virtual infrastructure modern networks run on, then turn to defending them with firewalls, access control and intrusion detection, finishing by writing your own secure networked applications at the socket level.

You'll explore:

- IP addressing, subnetting, routing and switching
- Virtualisation, hypervisors and virtual networks
- Firewalls, VPNs, IDS/IPS and secure network design
- Secure client-server programming with sockets

Level 7

Commercial Development Practice

Build real software for a real client with industry-standard project management techniques and commercial tools. In Commercial Development Practice, the School's own commercial software development unit, you'll work on genuine client projects that build the business and technical skills employers tell us are actively looking for.

You'll explore:

- Requirements analysis, defining specifications and system design
- Team-based implementation and quality control

- Client liaison, negotiation and project management
- Professional, reflective practice

Component Based Software Engineering Practice

From your phone to the server room, modern software runs on portable and reusable components — learn expert-level techniques to build the platforms that make that possible. Using .NET and the .NET CLR as your guide, you'll explore component-based programming environments and virtual machines, and are design, build and secure your own component-based system.

You'll explore:

- Versioning, metadata and reflection
- Virtual machine design and architecture
- Security in virtual machine execution
- Building a virtual machine

C++ Programming and Design

Core tools: C++, Assembly

Go deeper into C++ than most programmers ever will — right down to the processor. Across four core areas — language fundamentals, object design, memory management and low-level execution — you'll master the features that make C++ the language of choice for performance-critical software.

You'll explore:

- Inheritance, aggregation and virtual functions
- Smart pointers, dynamic memory and exception safety
- Assembly language and processor-level debugging
- Code optimisation techniques

Cutting Edge Concepts

The textbooks can't keep up with computer science — so this module looks at what's happening right now. You'll work with staff to explore genuinely current research across AI, games, security and beyond, learning to critically evaluate academic sources and separate rigorous research from the rest.

You'll explore:

- Identifying high-quality academic sources
- Critically reviewing current research
- Exploring topics beyond the core curriculum
- Skills that carry straight into postgraduate study or research careers

Dependable Reactive Real-Time Systems

Core tools: HiP-HOPS and SafeML for dependability analysis and understanding when machine-learning may no longer be trusted.

In avionics, autonomous vehicles and robotics, systems often have only one chance to get things right. This module looks at how complex software-intensive systems react and respond in real time, tolerate faults and remain safe when things go wrong.

You'll explore:

- Specification and modelling of reactive real-time systems
- Concurrency, communication and resource management
- Hazard identification and risk analysis
- Model-based dependability assessment.
- Fault tolerance, resilience and recovery
- **Real-time Graphics**

Core tools: Vulkan, GLSL

Learn to render in real time at the level used in AAA games and film. You'll master the modern graphics pipeline on real shader-programmable hardware, implementing the lighting, shading and rendering techniques used across the games and animation industries.

You'll explore:

- Parallel processing models in 3D graphics
- The real-time graphics pipeline, hands-on
- Lighting, shading, texturing and shadow generation
- Multi-pass rendering and deferred shading

Edge AI

AI doesn't just live in the cloud — learn to run it on the device, in real time, wherever it's needed. You'll design intelligent systems that work under real-world constraints of power, memory and connectivity, building and deploying AI models directly onto edge hardware and connecting them through IoT.

You'll explore:

- Real-time data processing on edge-capable hardware
- Lightweight AI models for devices like Raspberry Pi and Jetson
- IoT integration and sensor networks
- A full end-to-end edge AI project

Simulation and Concurrency

Core tools: C++, Compute Shaders, Sockets

Simulate real physics, run it in parallel, and share it across a network — all in one module. You'll build real-time physical simulations, explore concurrent programming across CPU and GPU, and implement networked, distributed solutions using sockets.

You'll explore:

- Newtonian mechanics and collision response
- Parallel programming with threading and GPU concurrency
- Lock-based and lockless synchronisation
- Networked, distributed application design

Trustworthy Computing

Security isn't an afterthought — it's a discipline. This module teaches you to build it in from the start, exploring the principles behind systems that are genuinely secure and private, from cryptography to the ethical and legal responsibilities that come with it.

You'll explore:

- Trust, authentication and security risk
- Cryptography and cryptographic algorithms
- The role of virtual machines in system security
- Securing distributed systems

Cyber Defence and Threat Intelligence

In today's rapidly evolving threat landscape, cybersecurity professionals must understand both how attacks are carried out and how organisations defend against them. This hands-on module combines offensive and defensive security perspectives, giving you the opportunity to develop the mindset and skills used by modern cyber defence professionals. Working across Red Team, Blue Team, and Purple Team scenarios, you'll learn how to detect threats, investigate incidents, analyse malicious activity, and strengthen organisational security.

You'll explore:

- Red Team, Blue Team, and Purple Team operations
- Vulnerability assessment techniques
- Enterprise security challenges, including Active Directory attacks and defences
- Digital forensics and malware investigation
- Threat intelligence collection, analysis, and reporting
- Industry-standard frameworks, methodologies, and best practices

Responsible, Safe and Explainable AI

The most powerful AI in the world is worthless if nobody can trust it — this module teaches you to build AI people can rely on. You'll explore the cutting edge of responsible, explainable AI, examining how fairness, safety, explainability and transparency can be built into systems from the ground up, not bolted on afterwards.

You'll explore:

- Explainable AI (XAI) and interpretable machine learning
- Detecting and mitigating bias in AI systems
- AI safety, robustness and adversarial attacks
- Responsible multi-agent and autonomous AI systems
- Federated and privacy-preserving learning
- Building AI that is fair, transparent, accountable and trustworthy